SUMMARY

Key findings at a glance

PLA/MWCNT/FDM tensile strength · strict set $n = 26$ · multi-route pool $n = 44$ · leave-one-out evaluation

Metric (90% level)	Bootstrap under-covers · overconfident	Bayesian NUTS over-covers · conservative
PICP (coverage)	0.846	1.000
MPIW, MPa (width)	25.9 sharper	28.9
NLPD	3.59	3.58 tied
CRPS	4.59	5.03
ECE	0.105	0.099 ~tied
Min. specimens (PICP ≥ 0.88)	5	5
Relative per-fit cost	1×	~350×

Under like-for-like LOO the two flat methods are **statistically equivalent on score** — they differ only in **failure mode** (bootstrap overconfident and sharp; Bayesian conservative and wide). The decisive Bayesian gain is **hierarchical**: partial pooling recovers a credible +0.166 CNT-loading slope for the $n = 10$ non-FDM group that bootstrap cannot fit, exposing a 2.2× FDM-vs-non-FDM route effect a flat model conceals.